LAB 7 – FUZZ TESTING

Fuzz Testing for Input Validation Bugs

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Subject / Lab	Software Engineering Lab 7

SCREENSHOT EVIDENCE

1. ORIGINAL APPLICATION EXECUTION (BEFORE FIX)

```
(.venv) PS C:\Users\DELL\Desktop\5th sem\SE LAB\SE-Labs-PES1UG24CS596\Lab 7>
 python fuel_station.py
==== Running parse_purchase ====
Enter grade:litres (e.g. diesel:25): diesel:25
Grade and Litres: ('Diesel', 25.0)

==== Running volume_average ====
Enter pump volumes in litres as CSV (e.g. 10,25,40): 10,25,40
Average Volume: 25

==== Running payable_bill ====
Enter bill|discount (e.g. 2500|100): 2500|100
Payable Bill: 2850.0
(.venv) PS C:\Users\DELL\Desktop\5th sem\SE LAB\SE-Labs-PES1UG24CS596\Lab 7>
```

2. HYPOTHESIS TESTS FAILING ON ORIGINAL CODE

```
(.venv) PS C:\Users\DELL\Desktop\5th sem\SE LAB\SE-Labs-PES1UG24CS596\Lab 7> python -m pytest test_fuel_station.py -v --tb=short
============================= test session starts =============================
platform win32 -- Python 3.13.14, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\DELL\Desktop\5th sem\SE LAB\SE-Labs-PES1UG24CS596\Lab 7\.venv\Scripts\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\DELL\Desktop\5th sem\SE LAB\SE-Labs-PES1UG24CS596\Lab 7
plugins: hypothesis-6.170.15
collected 6 items

test_fuel_station.py::test_parse_purchase_no_crash FAILED                [ 16%]
test_fuel_station.py::test_volume_average_no_crash FAILED                [ 33%]
test_fuel_station.py::test_payable_bill_no_crash FAILED                  [ 50%]
test_fuel_station.py::test_parse_purchase_valid PASSED                   [ 66%]
test_fuel_station.py::test_volume_average_valid PASSED                   [ 83%]
test_fuel_station.py::test_payable_bill_valid PASSED                     [100%]

================================== FAILURES ===================================
_____________________ test_parse_purchase_no_crash _____________________
```

3. CORRECTED APPLICATION EXECUTION (AFTER FIX)

```
(.venv) PS C:\Users\DELL\Desktop\5th sem\SE LAB\SE-Labs-PES1UG24CS596\Lab 7> python
 fuel_station.py
==== Running parse_purchase ====
Enter grade:litres (e.g. diesel:25): diesel:25
Grade and Litres: ('Diesel', 25.0)

==== Running volume_average ====
Enter pump volumes in litres as CSV (e.g. 10,25,40): 10,25,40
Average Volume: 25

==== Running payable_bill ====
Enter bill|discount (e.g. 2500|100): 2500|100
Payable Bill: 2850.0
(.venv) PS C:\Users\DELL\Desktop\5th sem\SE LAB\SE-Labs-PES1UG24CS596\Lab 7>
```

4. HYPOTHESIS TESTS PASSING AFTER FIX

```
(.venv) PS C:\Users\DELL\Desktop\5th sem\SE LAB\SE-Labs-PES1UG24CS596\Lab 7> python
 -m pytest test_fuel_station.py -v
============================= test session starts ==============================
platform win32 -- Python 3.13.14, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\DELL\Deskt
op\5th sem\SE LAB\SE-Labs-PES1UG24CS596\Lab 7\.venv\Scripts\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\DELL\Desktop\5th sem\SE LAB\SE-Labs-PES1UG24CS596\Lab 7
plugins: hypothesis-6.170.15
collected 6 items

test_fuel_station.py::test_parse_purchase_no_crash PASSED                [ 16%]
test_fuel_station.py::test_volume_average_no_crash PASSED                [ 33%]
test_fuel_station.py::test_payable_bill_no_crash PASSED                  [ 50%]
test_fuel_station.py::test_parse_purchase_valid PASSED                   [ 66%]
test_fuel_station.py::test_volume_average_valid PASSED                   [ 83%]
test_fuel_station.py::test_payable_bill_valid PASSED                     [100%]

============================== 6 passed in 1.46s ==============================
(.venv) PS C:\Users\DELL\Desktop\5th sem\SE LAB\SE-Labs-PES1UG24CS596\Lab 7>
```